



INDIA CRIME INTELLIGENCE DASHBOARD

Business Intelligence Project Report | 2001 – 2013

*An end-to-end Business Intelligence project analyzing Indian crime trends using
Microsoft Excel, Power BI, DAX, and Power Query*

Prepared by

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Tools: Microsoft Excel • Power BI • DAX • Power Query

1. Project Overview

The India Crime Intelligence Dashboard is an end-to-end Business Intelligence project that analyzes Indian crime data from 2001 to 2013 using Microsoft Excel and Power BI.

The project demonstrates the complete analytics workflow, including data cleaning, transformation, data modeling, DAX calculations, KPI development, and interactive dashboard design. It enables users to explore crime trends across States, Districts, Years, and Crime Categories while providing valuable insights into crimes against women, regional crime distribution, and year-over-year growth.

The solution combines the analytical capabilities of Excel with the visualization power of Power BI, transforming raw crime records into meaningful business intelligence for better decision-making.

2. Project Objectives

- Analyze crime trends across India from 2001 to 2013.
- Identify high-crime States and Districts.
- Compare different IPC crime categories.
- Analyze crimes against women separately.
- Perform year-over-year crime growth analysis.
- Identify regional crime hotspots.
- Build an interactive multi-page dashboard for data-driven decision making.
- Demonstrate an end-to-end Business Intelligence workflow using Excel and Power BI.

3. Tools & Technologies Used

- Microsoft Excel
- Microsoft Power BI
- DAX (Data Analysis Expressions)
- Power Query
- Pivot Tables
- Charts & Visualizations
- Conditional Formatting

4. Data Cleaning

Before building the dashboards, the dataset was cleaned and standardized to improve accuracy and consistency. The following steps were carried out:

- Removed duplicate records.
- Standardized inconsistent State and District names.
- Corrected naming inconsistencies such as A&N Islands and D&N Islands.
- Removed unnecessary aggregate rows (ZZ Total, Delhi UT Total).
- Corrected formatting inconsistencies.
- Validated numerical values.

- Verified missing values.
- Standardized the overall dataset structure.
- Prepared the dataset for Power BI data modeling.

5. Dashboard Pages

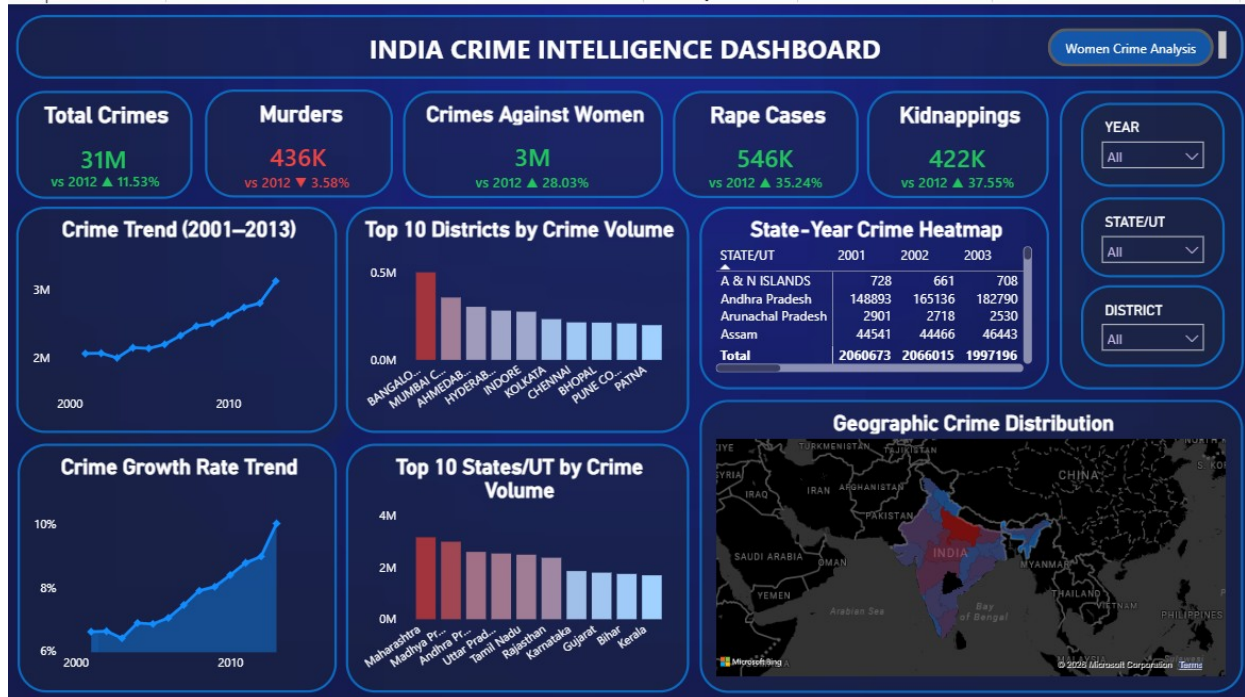
5.1 Dashboard 1 — National Crime Overview

Key Performance Indicators:

• Total Crimes
• Total Murder Cases
• Total Crimes Against Women
• Total Rape Cases
• Total Kidnapping Cases

Key Visuals:

- Crime Trend Analysis
- Top 10 States by Crime
- Top 10 Districts by Crime
- Year-over-Year Growth Analysis
- Dynamic KPI Cards
- Interactive Slicers



Dashboard 1 – National Crime Overview

5.2 Dashboard 2 — Women Crime Intelligence

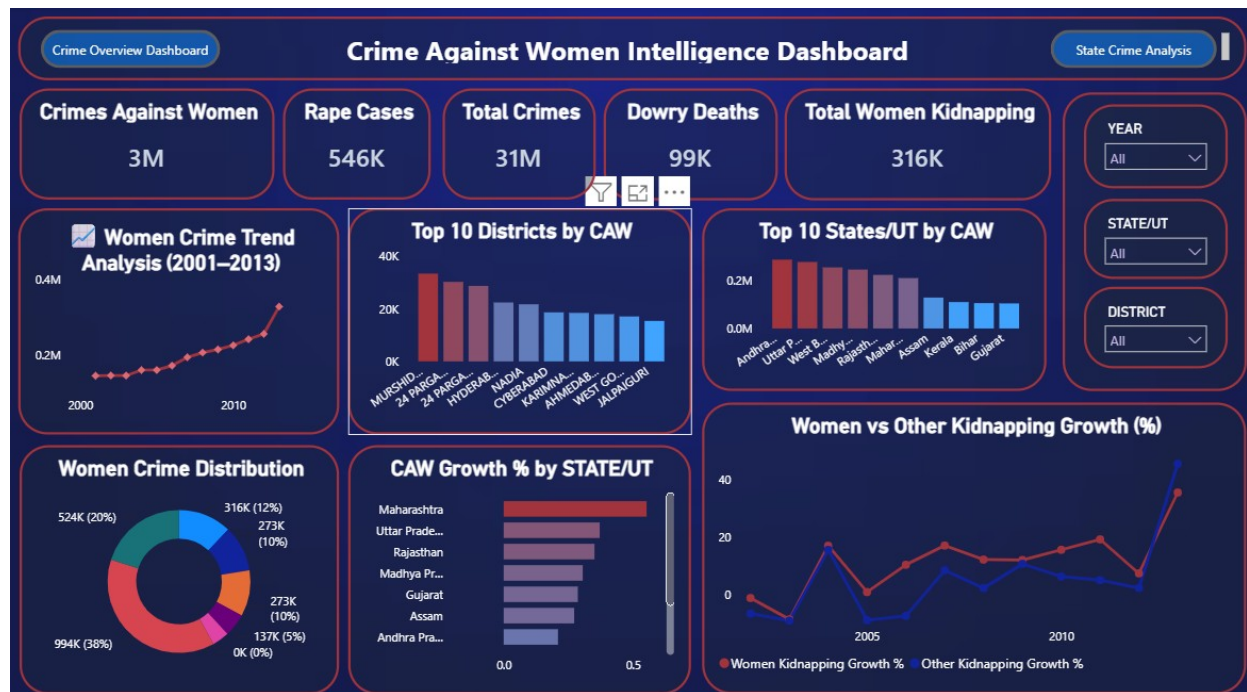
Key Performance Indicators:

• Total Crimes Against Women

- Total Rape Cases
- Women Kidnapping Cases
- Dowry Deaths
- Women Crime Growth %

Key Visuals:

- State-wise Women Crime Analysis
- District-wise Women Crime Analysis
- Women vs Other Kidnapping Comparison
- Yearly Crime Trend
- Interactive KPI Cards



Dashboard 2 – Women Crime Intelligence

5.3 Dashboard 3 — State Crime Intelligence**Key Performance Indicators:**

- States Covered
- Top Crime State
- Peak Crime Year
- Peak Crime Volume
- Average Crime per State

Key Visuals:

- Top 10 States by Crime
- Crime Hotspot Map
- State-Year Heatmap
- Top 5 Crime Categories
- Crime Distribution Analysis

- Dynamic Crime Type Comparison



Dashboard 3 – State Crime Intelligence

5.4 Dashboard 4 — District Crime Intelligence

Key Performance Indicators:

• Districts Covered
• Top Crime District
• Peak Crime District
• Peak Crime Volume
• Average Crime per District

Key Visuals:

- Top District Analysis
- District Crime Trend
- Crime Distribution
- District Heatmap
- Drill-through Analysis



Dashboard 4 – District Crime Intelligence

6. Power BI Features Implemented

- Dynamic KPI Cards
- Dynamic Titles
- Dynamic Tooltips
- Advanced DAX Measures
- Year-over-Year Analysis
- Interactive Slicers
- Cross Filtering
- Navigation Buttons
- Drill-through Pages
- Conditional Formatting
- Filled Maps
- Heatmaps
- Responsive Dashboard Design
- Dynamic Crime Type Selection

7. Key Insights

- Maharashtra records the highest overall crime volume across the study period.
- Uttar Pradesh consistently reports one of the highest murder case counts.
- Theft is the most frequently reported IPC crime category.
- Crimes against women show a continuous increasing trend over the years.
- Women kidnapping cases have increased significantly during the study period.
- The year 2013 recorded the highest overall crime volume.
- Dacoity cases have gradually declined, while Theft and Robbery continue to rise.

- Highly populated states contribute a larger share of national crime.
- District-level analysis highlights concentrated crime hotspots requiring targeted interventions.
- Interactive dashboards enable dynamic analysis by State, District, Year, and Crime Category.

8. Recommendations

- Strengthen law enforcement in high-crime states.
- Increase investment in women safety initiatives and awareness programs.
- Monitor district-level crime trends for targeted policing.
- Use year-over-year analysis to identify emerging crime hotspots.
- Allocate policing resources based on crime intensity.
- Focus preventive measures on frequently occurring crime categories.
- Utilize interactive dashboards for faster and evidence-based decision making.

9. Outcome

This project successfully transformed raw crime records into an interactive Business Intelligence solution that provides comprehensive insights into India's crime patterns. The dashboards enable users to explore crime trends across multiple dimensions, identify regional hotspots, compare crime categories, and analyze year-over-year changes. The project demonstrates practical expertise in Excel, Power BI, DAX, data visualization, and business intelligence, while showcasing the ability to convert raw data into meaningful and actionable insights.

10. Project Highlights

- End-to-End Data Analytics Project
- Microsoft Excel + Power BI Integration
- Interactive 4-Page Dashboard
- Advanced DAX Calculations
- Dynamic KPI Cards
- Interactive Slicers
- Dynamic Tooltips
- Year-over-Year Analysis
- Crime Hotspot Analysis
- Interactive Maps
- Heatmaps
- Drill-through Navigation
- Professional Dashboard Design
- Business Intelligence Reporting

11. Files Included

The repository contains all project files required to understand, reproduce, and explore the complete analysis:

- Excel File (Raw Data)
- Excel Dashboard
- One Page Summary (Excel)

- Project Report (PDF)
- Excel Dashboard Screenshot (PNG)
- Power BI Dashboard (.pbix)

12. About the Author

Prakhar Kayat — Aspiring Data Analyst

Skills:

- Microsoft Power BI
- Microsoft Excel
- SQL, PostgreSQL
- DAX
- Data Visualization
- Python (NumPy, Pandas, Matplotlib, Seaborn)